

ORE TEXTURES

GLM-205

INTRODUCTION

- **The spatial relationships between grains.**

Why texture is important?

```
graph TD; A[Why texture is important?] --> B[Process during ore formation]; A --> C[Process after ore formation];
```

Process during ore formation

Process after ore formation

Refractoriness of ore minerals

- **Oxides,**
- **Disulfides,**
- **Arsenides, and**
- **Sphalerite**

- **Pyrrhotites or Cu-Fe sulfides**

- **Pyrrhotites or Cu-Fe sulfides.**
- **Argentite, sulfosalts, and native metals**

Most Refractory

Least Refractory

Broad division of ore textures

Primary texture: are formed during the process of ore formation

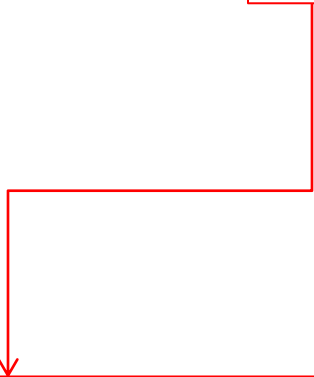
Secondary texture: are formed when ore is subjected to later processes (metamorphism, deformation, annealing, and meteoric weathering)

Broad division of ore textures

Primary texture: are formed during the process of ore formation

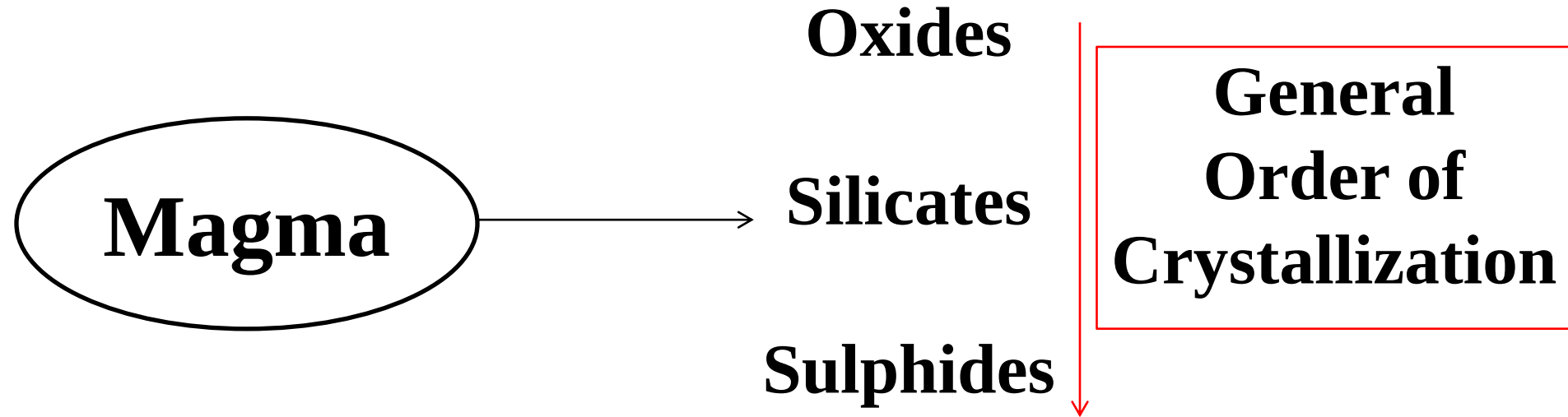
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Primary Texture

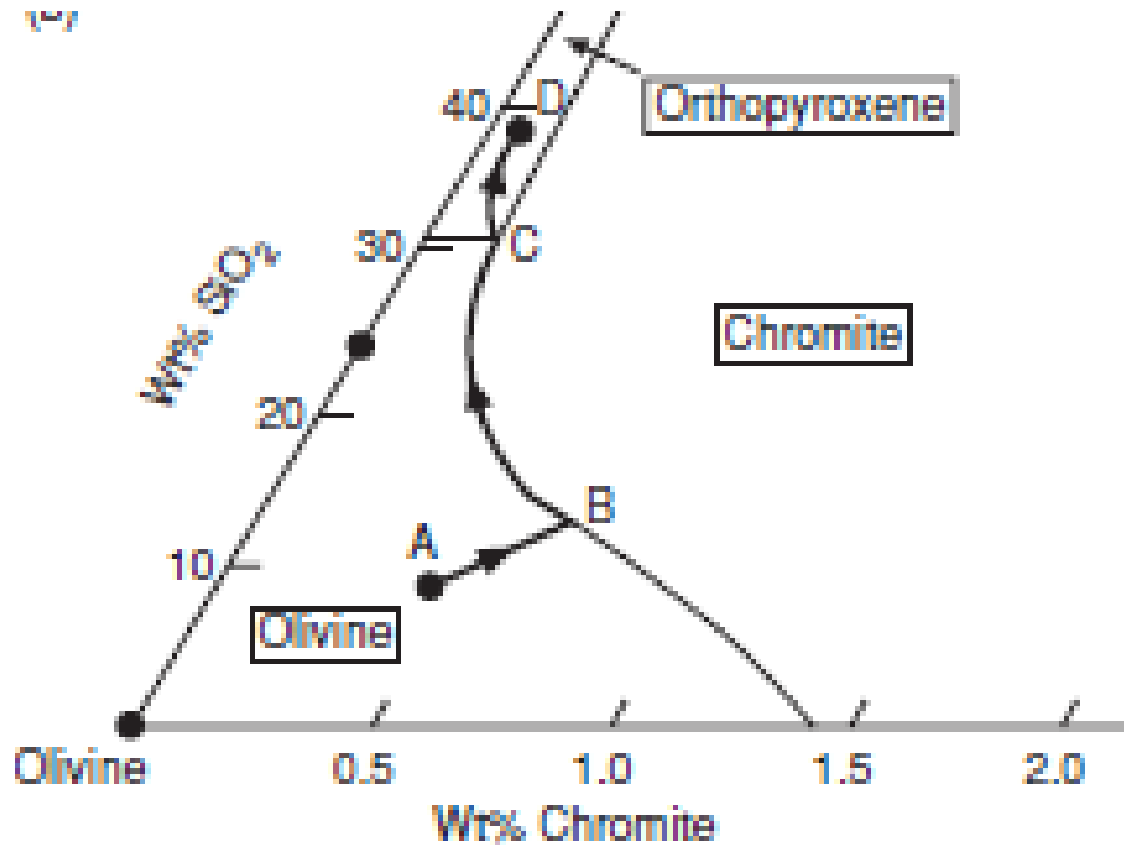


- **Primary textures of ore minerals formed from melt**
- **Primary textures of open space deposition**

Primary textures of ore minerals formed from melt

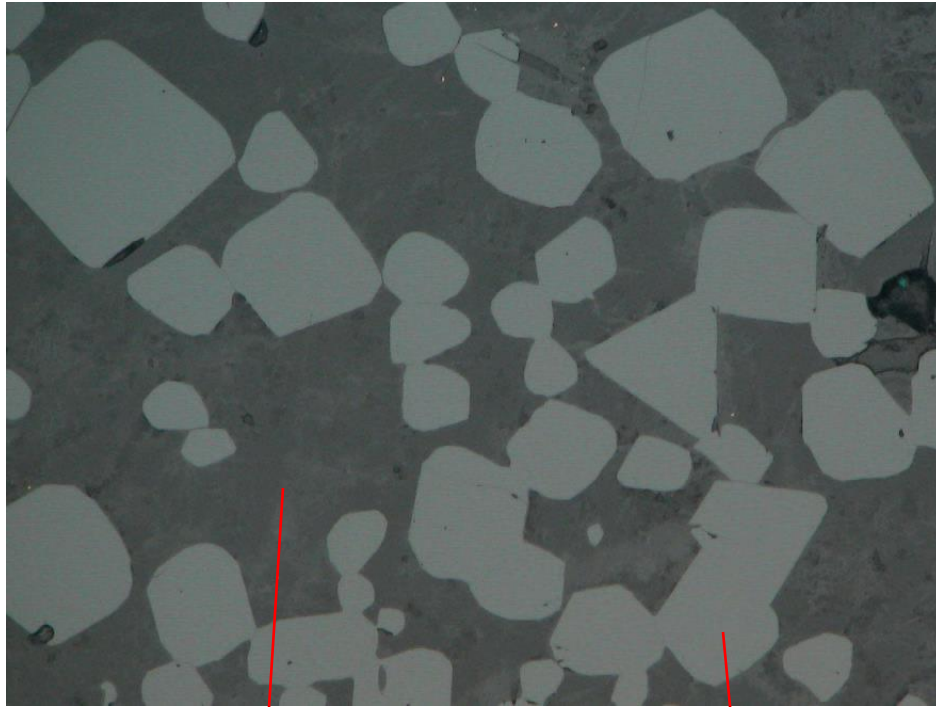


Primary textures of ore minerals formed from melt



The nature of crystallization in a mafic magma explained in Irvine's Model (Robb, 2005)

Primary textures of ore minerals formed from melt

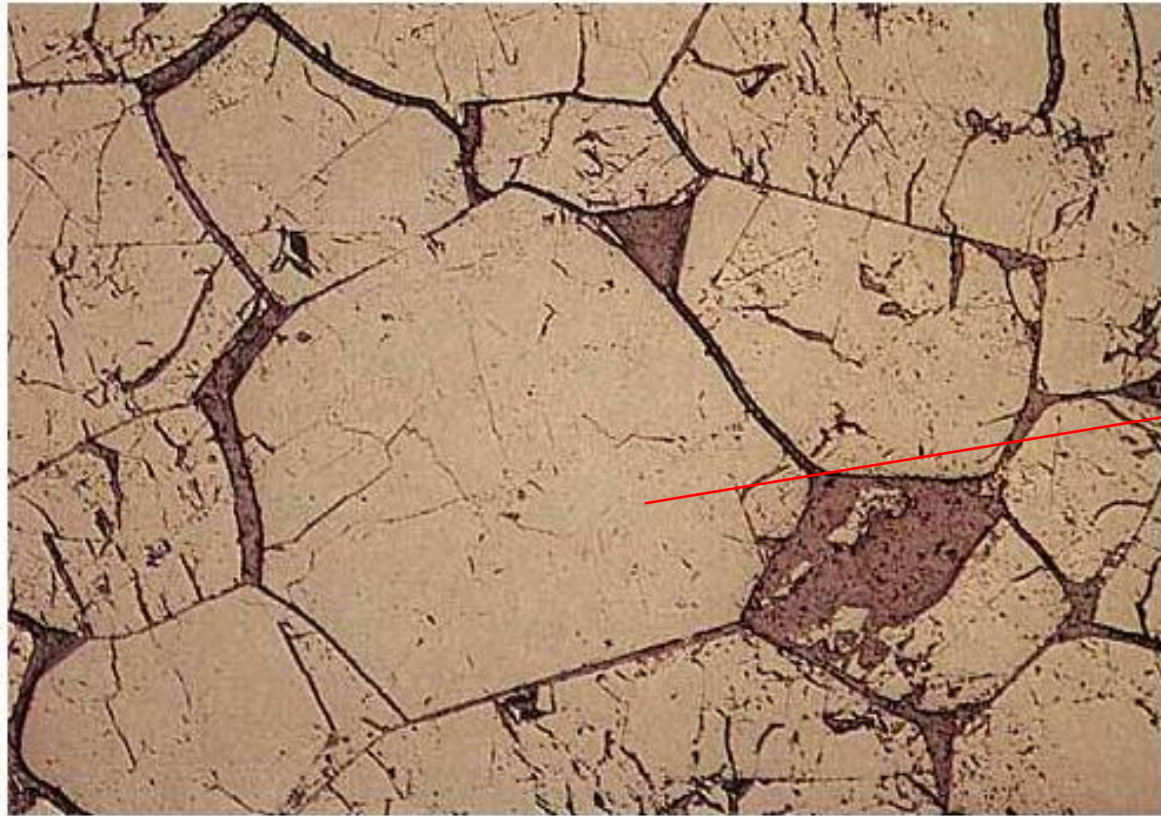


Euhedral Chromite grains in silicate matrix

Silicate

Chromite

Primary textures of ore minerals formed from melt

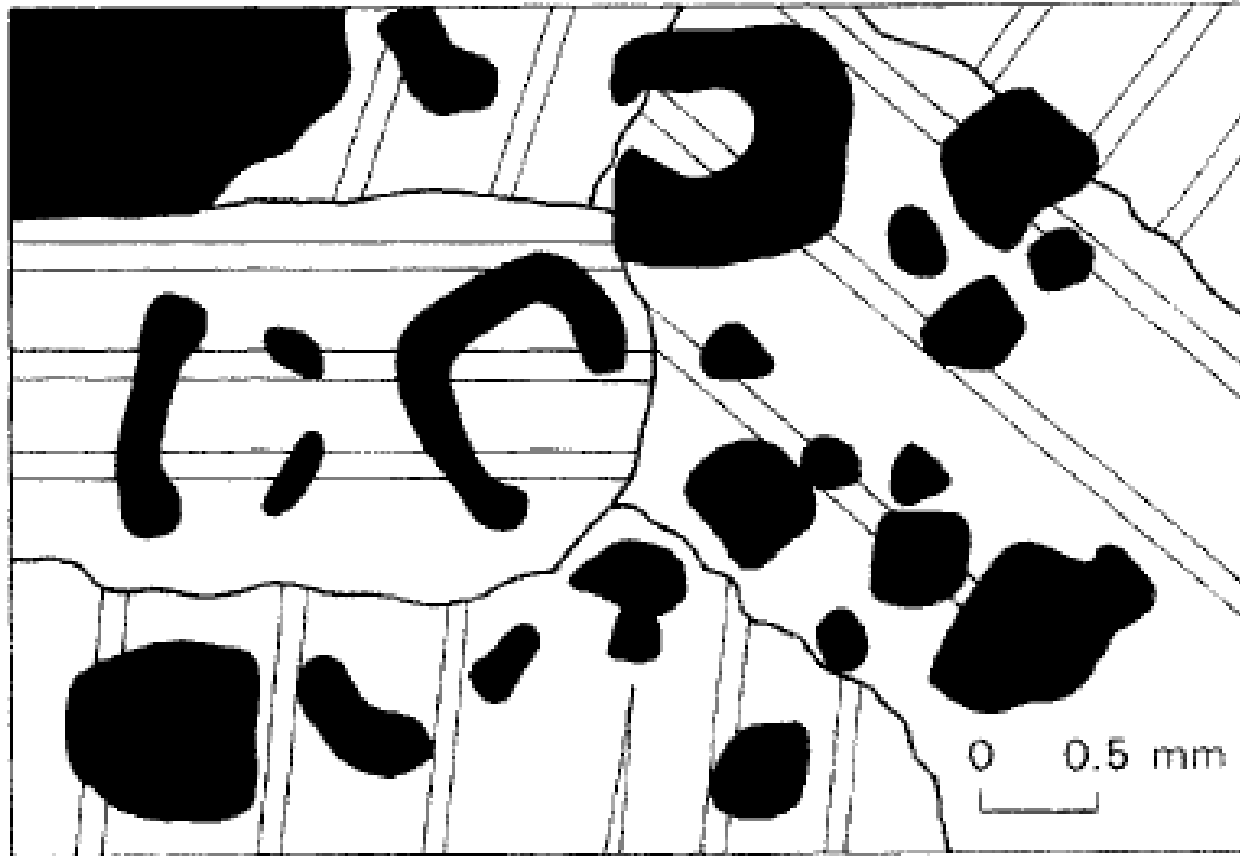


Chromite

125µm

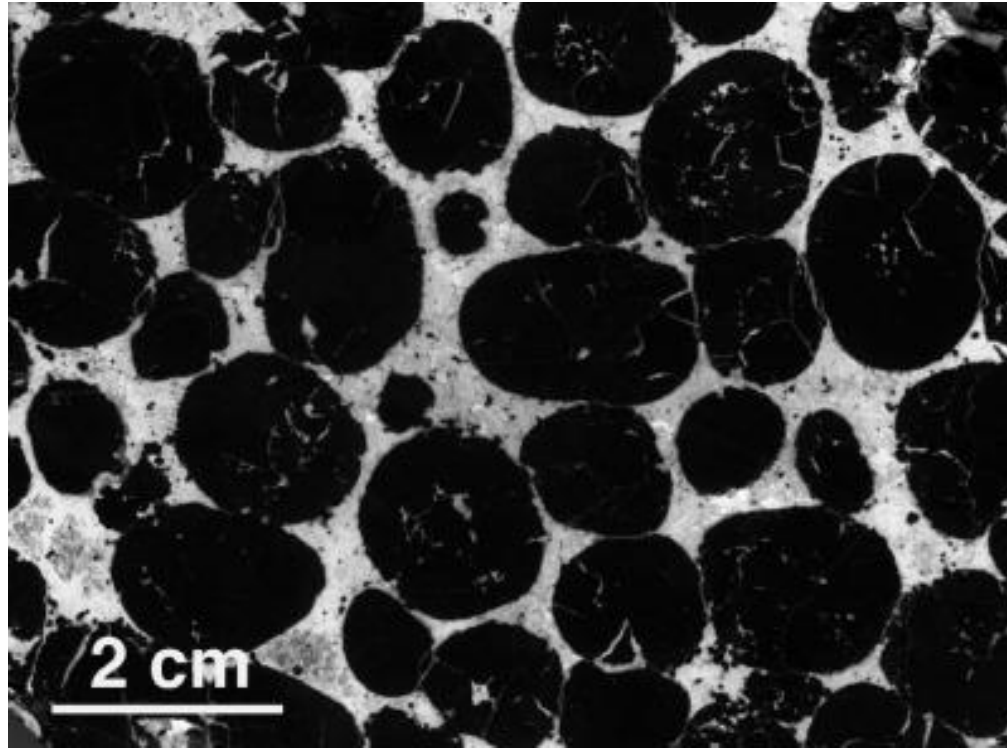
Chromite grains in a chromite rich layer

Primary textures of ore minerals formed from melt



Sketch of chromite grains in anorthosite, Bushveld Complex, RSA. The chromites are euhedral crystals which have undergone partial resorption, producing rounded grains of various shapes, including atoll texture.

Primary textures of ore minerals formed from melt



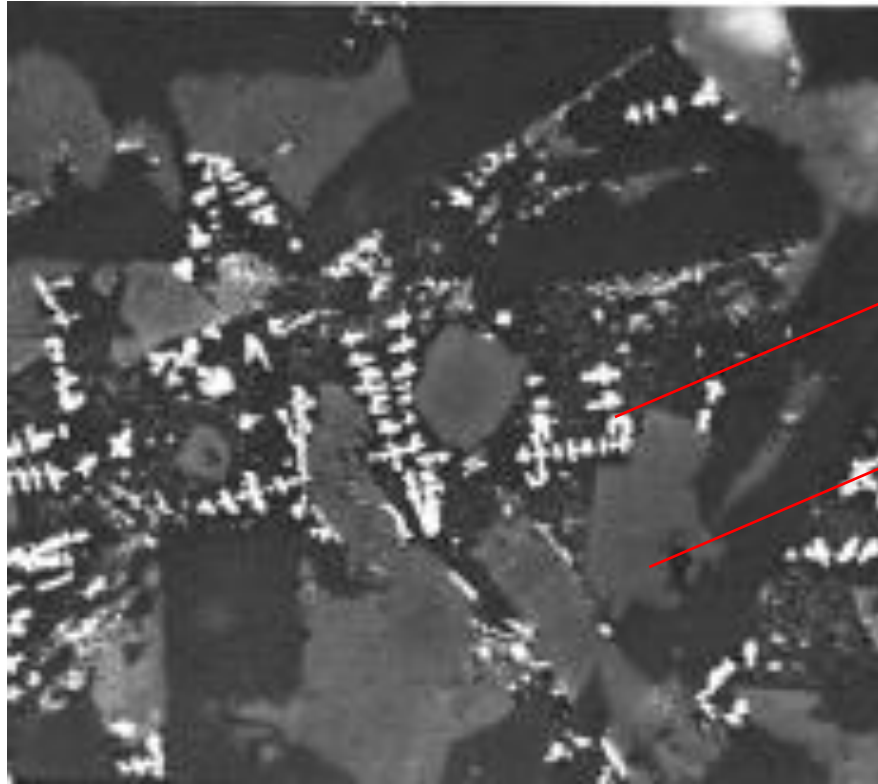
Possible Reason

- The nodules formed earlier and deformed or abraded by later movement of semi-solid mass

Orbicular grains of Chromite in case of Podiform Deposits

Primary textures of ore minerals formed from melt

What happens in case of rapidly cooled basalt?



Ilmenite

Silicate

Unobstructed growth, especially in rapidly cooled basalts, sometimes results in the formation of skeletal crystals that may be wholly or partially contained within subsequently solidified glasses or crystallizing silicates.

Primary textures of ore minerals formed from melt

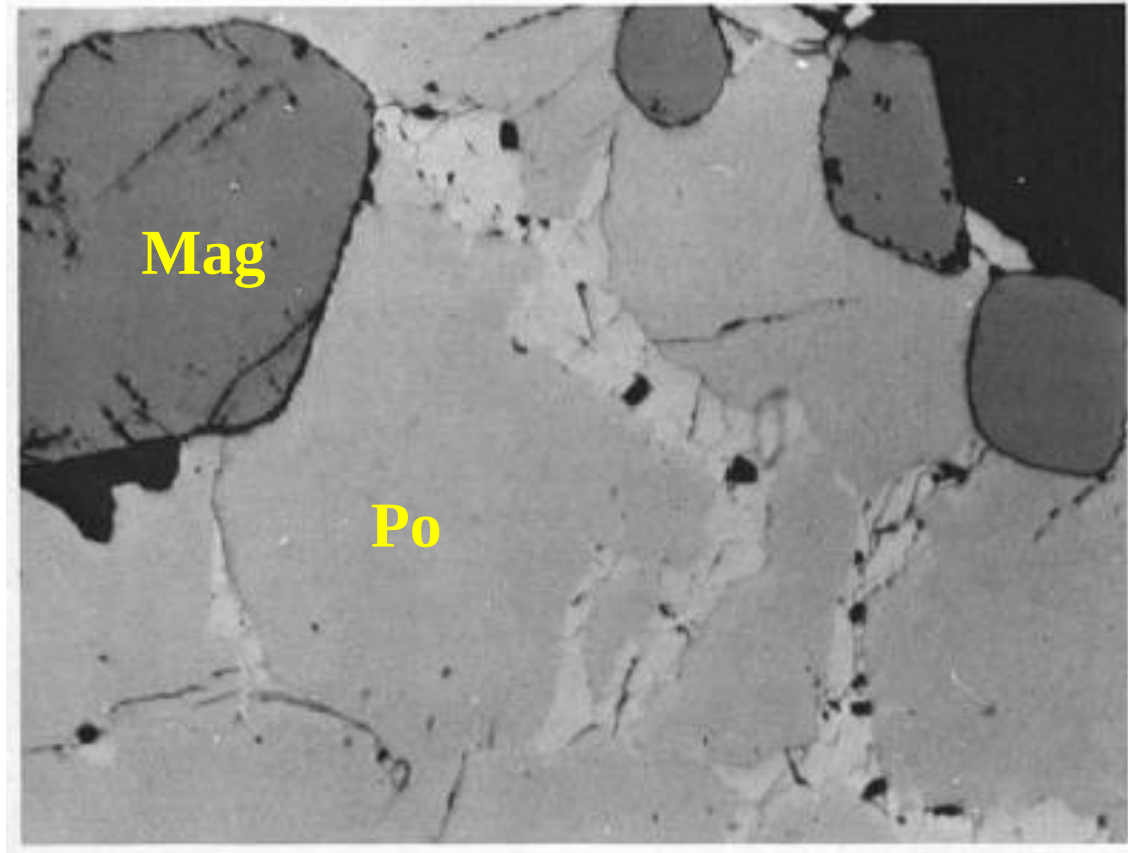
Textures Iron-Nickel-Copper Ore (Magmatic)

Melt Composition

Fe-Ni-Cu-S (-Oxygen)

Major Minerals that can be seen in Polished Section

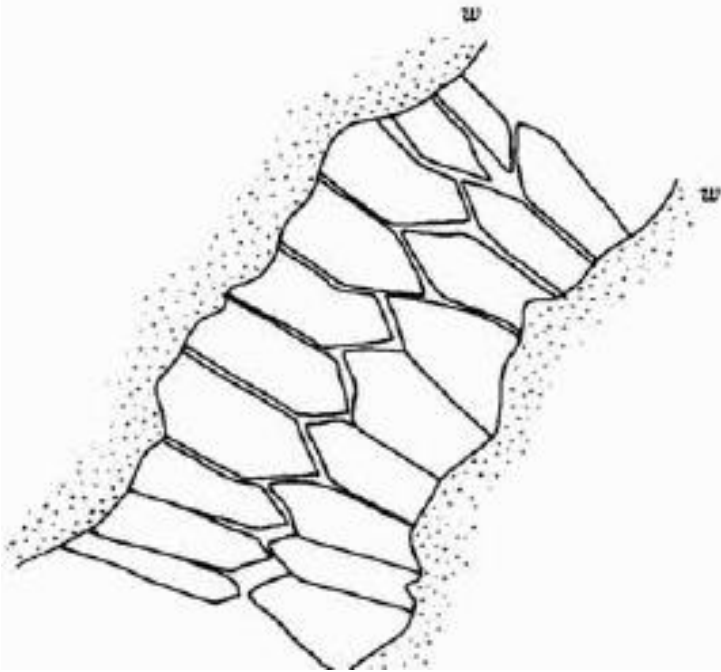
- Pyrrhotite
- Chalcopyrite
- Pentlandite
- Magnetite



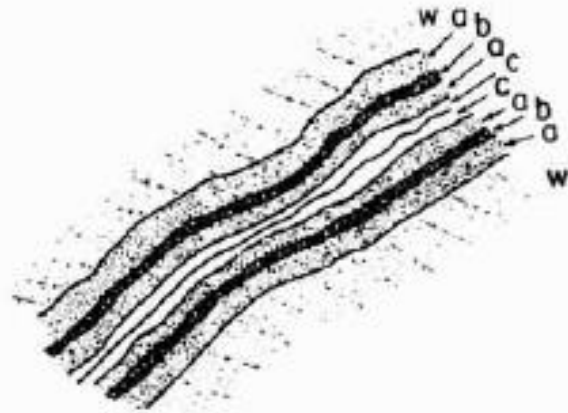
Primary iron-sulfur (-oxygen) melts can also result in the formation of small round droplets trapped in rapidly cooled basalts and basaltic glasses

Primary textures of open space deposition

Deposition in open spaces (cavities) generally characterised by mineral grains having well developed crystal faces due to unobstructed growth of mineral grains.



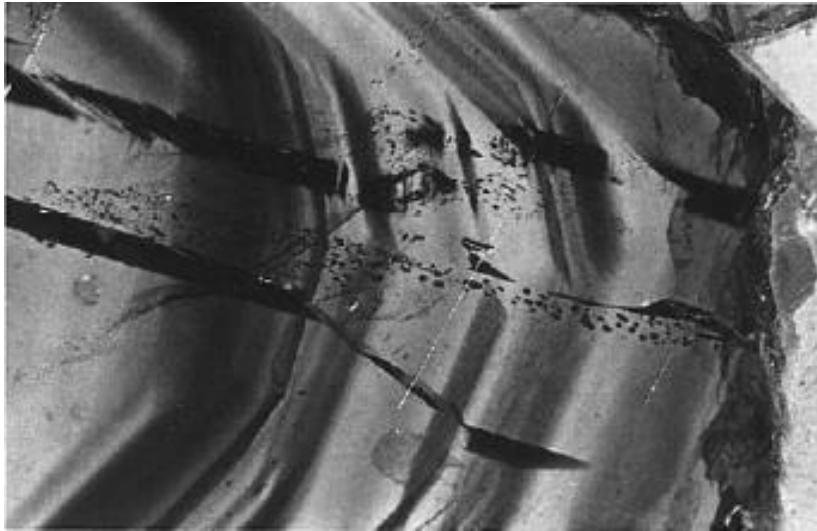
Comb Structure



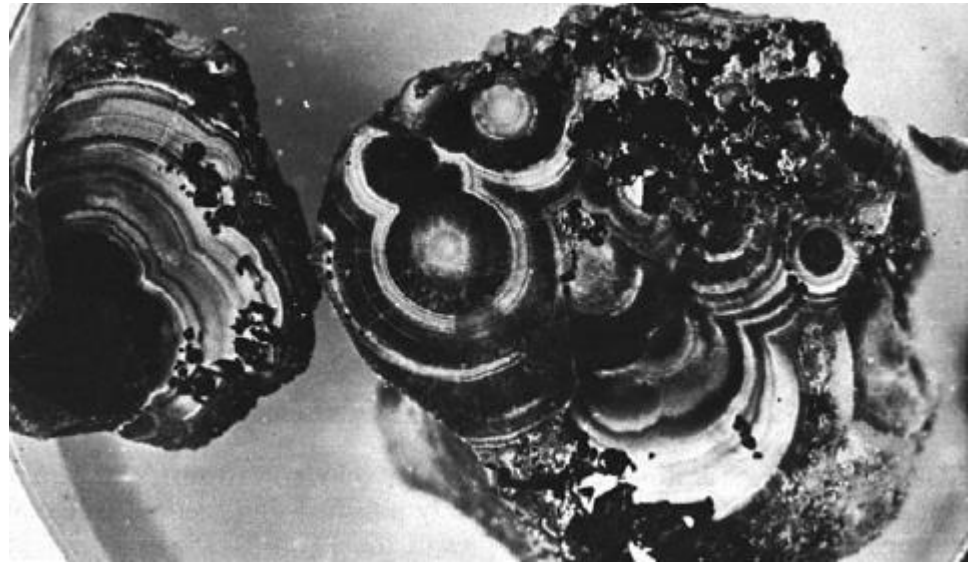
Crustification

Primary textures of open space deposition

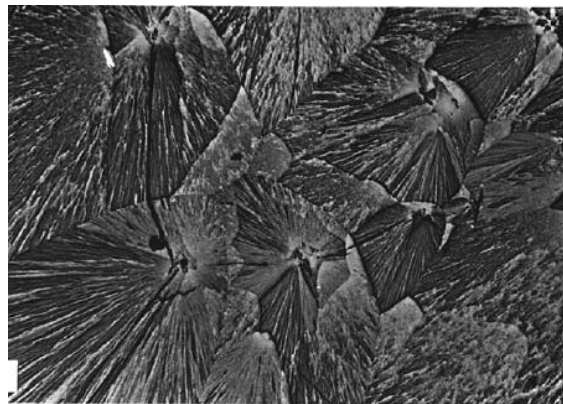
Growth Zoning and Colloform (Zoned monominerallic) bands



Growth zoning in a single crystal of sphalerite



Colloform growth banding in sphalerite



Radiating clusters of fibrous goethite crystals

Primary textures of open space deposition

Colloform Texture

Under **surficial conditions**, ore minerals may be deposited from **colloidal solutions**. A colloid is defined as a system consisting of two phases; one diffused in the other. Colloidal particles range in size between ions in a true solution and particles that are $< 10^{-3}$ cm in a coarse suspension. The colloidal material may be solid, liquid or gas dispersed in another solid, liquid or gas.

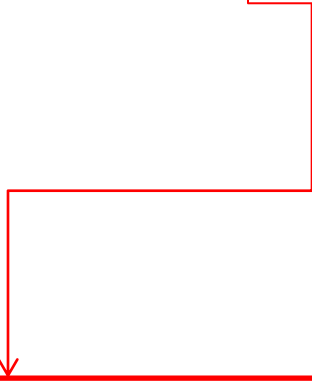
Colloidal solutions believed to be responsible for the formation of ore deposits usually consist of solids dispersed in liquids and are called "sols". In such sols, colloidal particles commonly adsorb either cations or anions, and thus acquire similar charges which cause them to repel each other, preventing them from coagulation. If an electrolyte is added to such a sol, the colloidal particles are neutralized and flocculate giving rise to a variety of textures which include:

- Botryoidal or reniform aggregates

- Banding or very fine layering

These textures are broadly lumped as "**colloform**" textures

Secondary Texture



```
graph TD; A[Secondary Texture] --> B[• Replacement<br/>• Cooling<br/>• Deformation<br/>• Annealing and Metamorphism];
```

- **Replacement**
- **Cooling**
- **Deformation**
- **Annealing and Metamorphism**

Secondary textures due to replacement

- **Replacements is due to interaction between a primary minerals and reacting fluid**
- **Difficult to establish replacement, unless the vestiges of primary mineral is present**

The major processes responsible for replacement are:

Dissolution

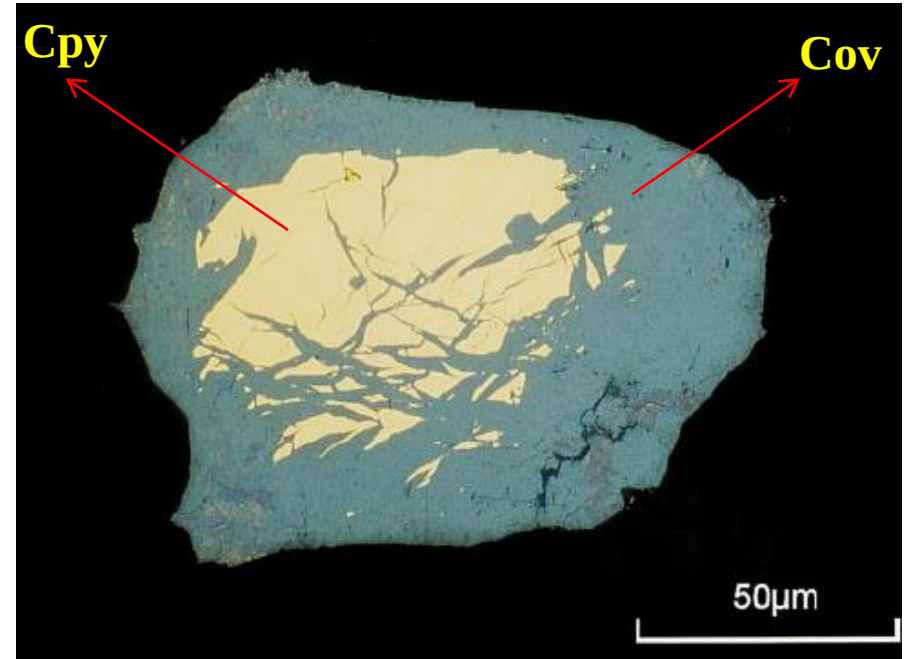
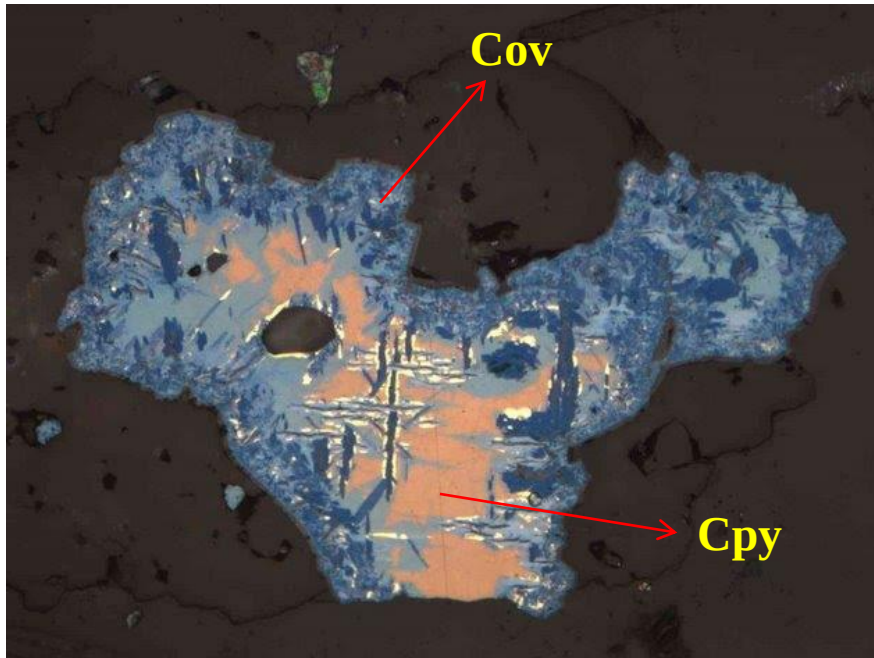
Oxidation

Solid state diffusion

Secondary textures due to replacement

Replacement textures depends on:

(i) The available surfaces for reaction: Grain boundaries and fractures are major sites for replacement

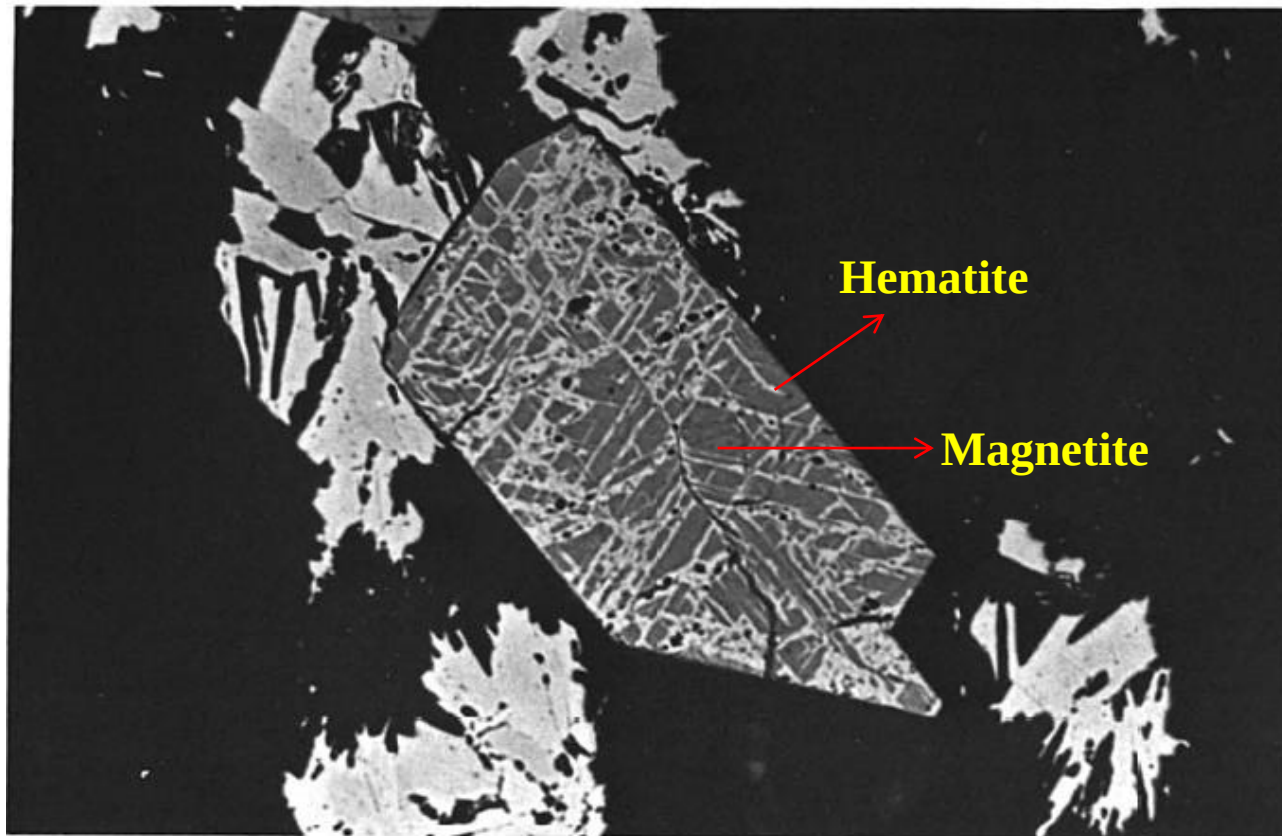


Replacement of chalcopyrite by covellite along grain boundaries and fractures. Cpy-Chalcopyrite Cov-Covellite

Secondary textures due to replacement

Replacement textures depends on:

(ii) Crystal Structure: Cleavage directions or along certain crystallographic directions

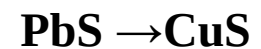
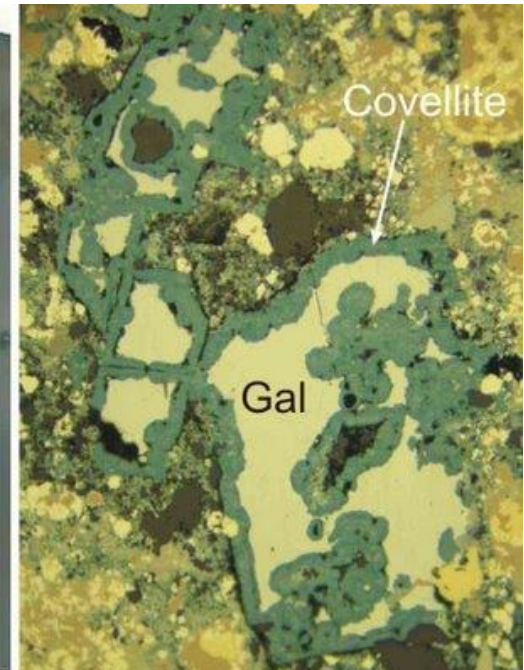
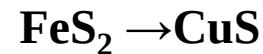
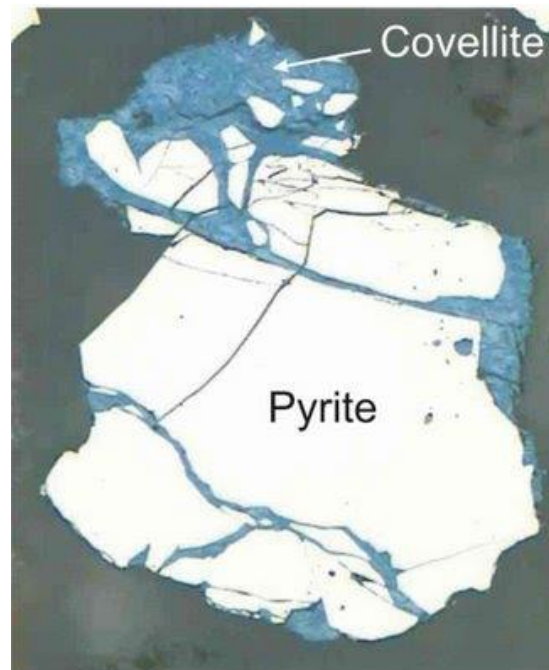
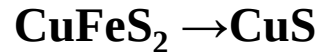
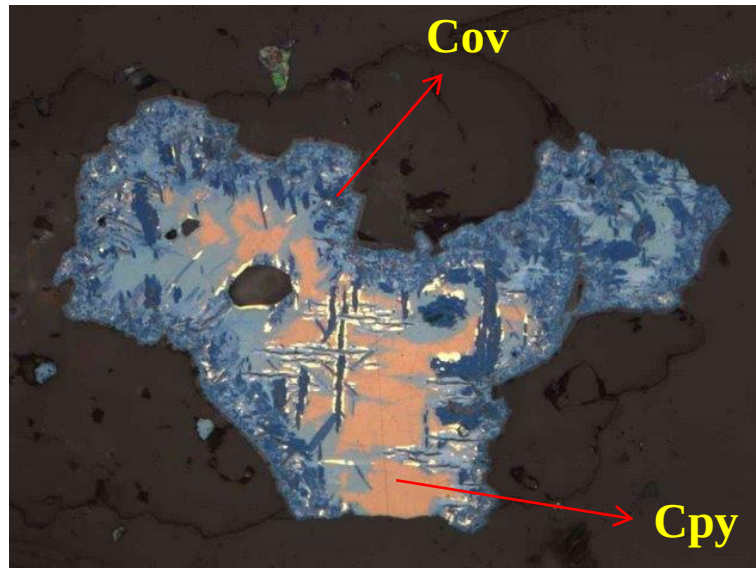


Martite: Replacement of magnetite by hematite along crystallographic preferred planes

Secondary textures due to replacement

Replacement textures depends on:

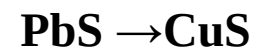
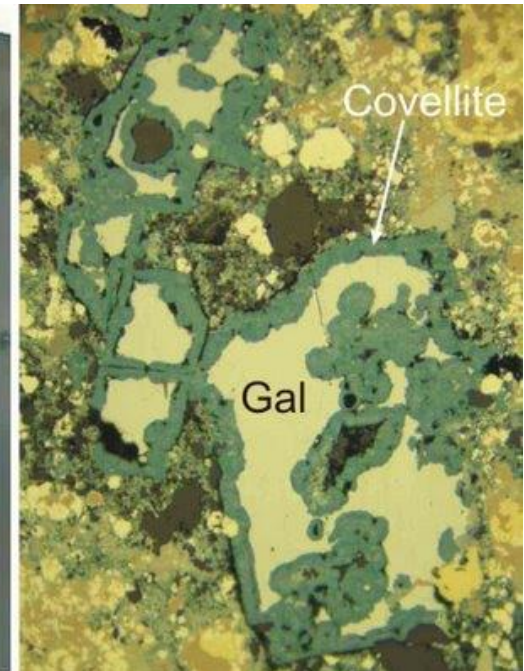
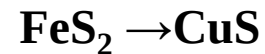
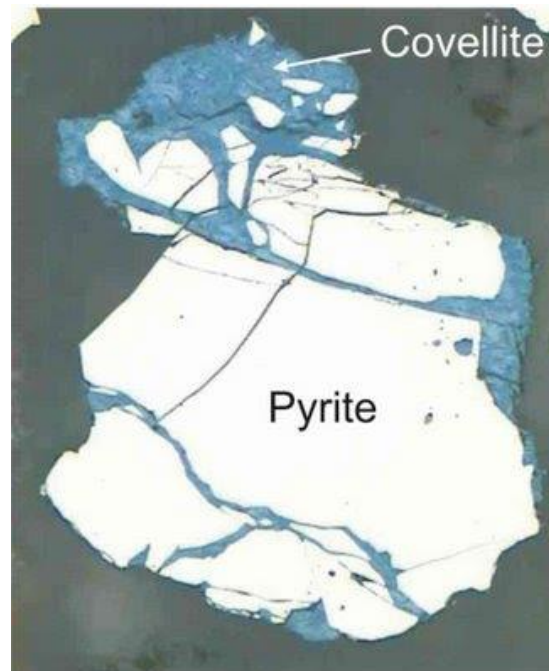
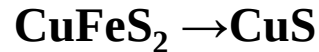
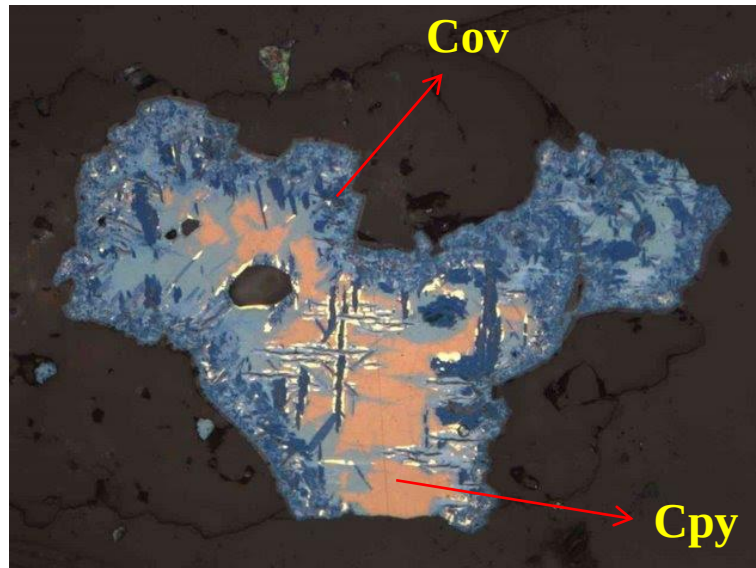
(iii) Chemical Composition of primary phase and reacting fluid



Secondary textures due to replacement

Replacement textures depends on:

(iii) Chemical Composition of primary phase and reacting fluid



Secondary textures due to cooling

Exsolution: One phase is expelled from another due to cooling

- Diffusion of atoms through a crystal
- Nucleation and growth of phase

Host Phase	Exsolved Phase	Nature of Commonly Observed Intergrowth
Arsenic or antimony	Stibarsen	Myrmekitic
Bornite	Chalcopyrite	Basket weave
Bornite	Chalcocite-digenite	Roughly cubic network
Chalcopyrite	Cubanite	Sharply bounded laths
Chalcopyrite	Sphalerite	Stars, crosses
Chalcopyrite	Mackinawite	Lamellae, irregular wisps
Chalcopyrite	Bornite	Basket weave
Galena	Matildite	Lamellae
Hematite	Ilmenite	Lens-like lamellae
Ilmenite	Hematite	Lens-like lamellae
Kamacite	Plessite	Lamellae in triangular pattern
Magnetite	Ilmenite	Lamellae in triangular pattern
Magnetite	Ulvöspinel	Lamellae in triangular pattern
Pb-Sb and Pb-Bi sulfosalts	Pb-Sb and Pb-Bi sulfosalts	Lamellar
Pyrrhotite	Pentlandite	Lamellae or flames
Silver	Dyscrasite	Lamellae
Sphalerite	Chalcopyrite	Rows of blebs
Sphalerite	Pyrrhotite	Rows of blebs
Sphalerite	Stannite	Dispersed blebs
Stannite	Chalcopyrite	Lamellae in triangular pattern
Stibarsen	Arsenic or antimony	Myrmekitic

List of ore minerals that frequently exhibit exsolution textures

Secondary textures due to cooling

Exsolution

The form of exsolved phase depends on

Mineral involved

Relative proportion of the phase

Cooling history

Secondary textures due to cooling

Exsolution

- a) Similarity of crystal structure and chemical bonding between host and exsolved phase along certain planes dictate that exsolution is crystallographically controlled. (Also known as coherent exsolution)

Example: Ulvospinel exsolves parallel to (111) planes of host magnetite

- b) If host and exsolved phases have different structures or if there is no crystallographical continuity across the interface between phases noncoherent exsolution occurs

Example: Pyrrhotite and Pyrite

Secondary textures due to cooling

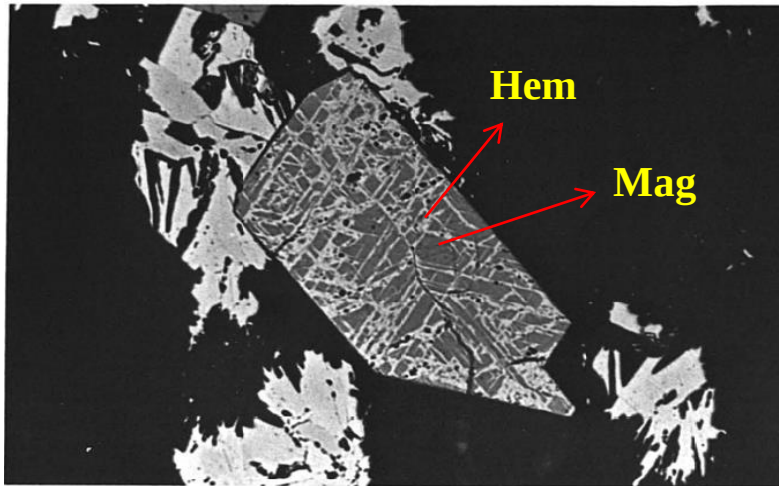
Exsolution

Terms used to describe the textures

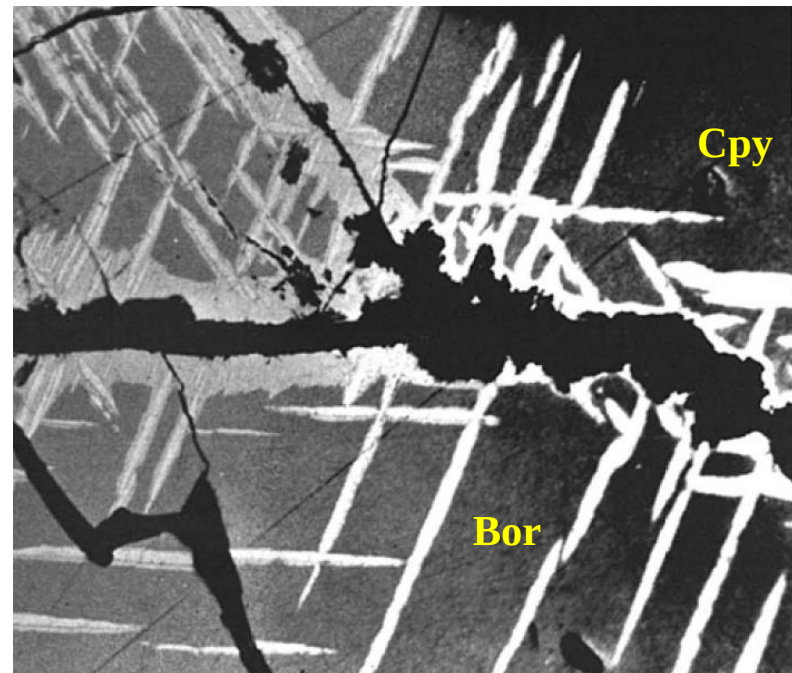
Marginal

Lamellar

Myermekite etc...



Alteration along crystallographic direction



Coherent Exsolution

Hem-Hematite, Mag-Magnetite, Cpy-Chalcopyrite, Bor-Bornite

Secondary textures due to cooling

Other Textures

- **Inversion**

Inversion of a mineral from one structural form to another of the same composition. But, hard to establish

- **Thermal Stress**

Fracturing of lamellae of pentlandite due to differential shrinkage (coefficient of thermal expansion)

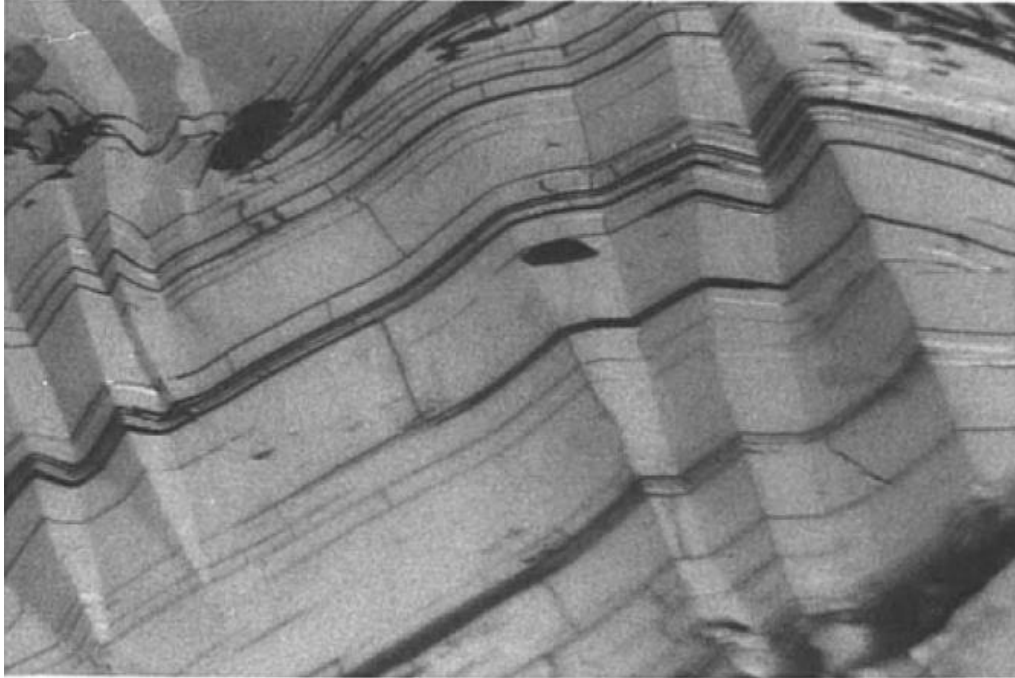
Secondary textures due to Deformation

- **Twining**

Growth	Occurs as lamellar twins of irregular width that are unevenly distributed, present in only some grains, and may be strongly interwoven
Inversion	Commonly occurs as spindle-shaped and intergrown networks not parallel throughout grains
Deformation	Occurs as uniformly thick lamellae, commonly associated with bending, cataclasis, and incipient recrystallization (regions of very small equant grains), with lamellae often passing through adjacent grains

Secondary textures due to Deformation

- **Kinkbands**



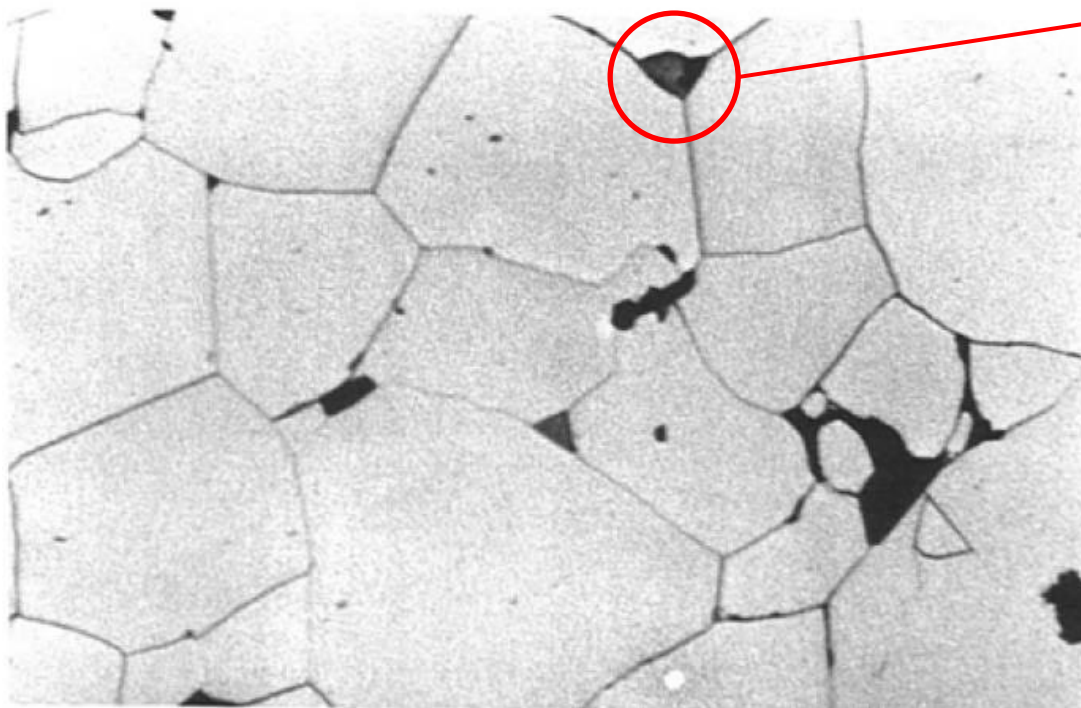
Kinkbanding in pyrrhotite accentuated by thin layers of micas

- **Curvature or offset of linear features**
- **Schlieren**
- **Brecciation**

Secondary textures due to Annealing and Metamorphic crystal growth

Annealing: Slow cooling of ore after deposition or slow heating during metamorphism

Minimization of grain surfaces and formation of equant grains



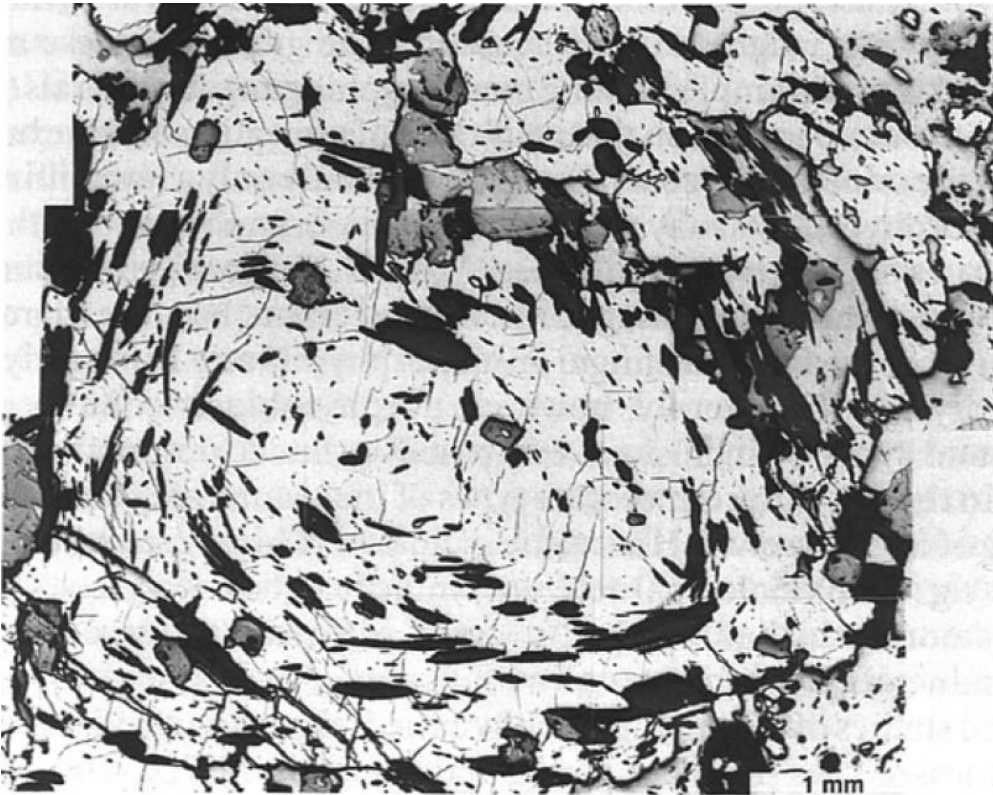
Interfacial angle at triple junction= 120°

Annealed texture of recrystallized monomineralic pyrite sample. Note the common development of near 120° triple junctions

Secondary textures due to Annealing and Metamorphic crystal growth

Metamorphism

- Increase in grain size
- Development of euhedral grains and sometimes porphyroblasts of pyrite, arsenopyrite, magnetite etc...



Pyrite crystal displaying a concentric pattern of amphibole inclusions incorporated during growth